Session3: Longitudinal Data Analysis

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Outline

1. Visualization: A way to exploratory analysis
2. Modeling: A way to confirmatory analysis

Introduction

Longitudinal data can be viewed as a special case of the multilevel data where time is nested within individual participants. All longitudinal data share at least three features:

(1) the same entities are repeatedly observed over time

(2) the same measurements (including parallel tests) are used

(3) the timing for each measurement is known

With longitudinal data, an obvious first graph is the scatterplot of the response variable against time. We use several references in addition to the main book. See:

Diggle, P.J., Heagerty, P., Liang, K.Y. and Zeger, S.L. (2002) Analysis of Longitudinal Data, 2nd Ed., Oxford University Press.

Summary

In longitudinal studies, people are measured repeatedly over time. This is in contrast to cross-sectional studies where a single outcome is measured for each individual.

: We can trace age effect (changes over time within individuals)

: We can capture cohort effects (subject effects) using differences among subjects in their baseline levels

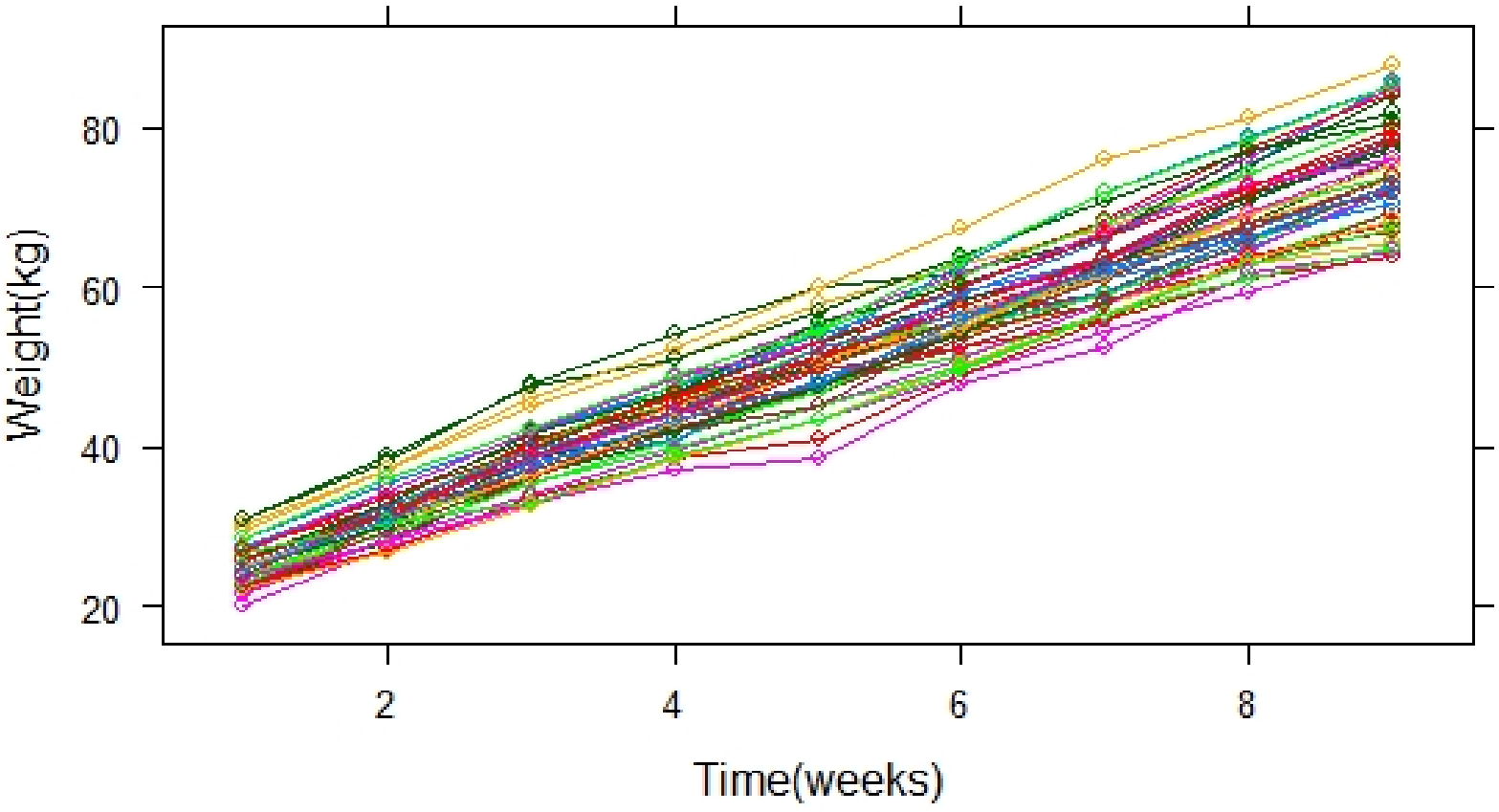
Because of the repeated measures over time, observations tend to be autocorrelated ($\rho(u) = \mathrm{corr}(Y_{ij}, Y_{ij-u})$) and thus we need to take of the covariance/correlation in modeling longitudinal data.

Graphical presentation

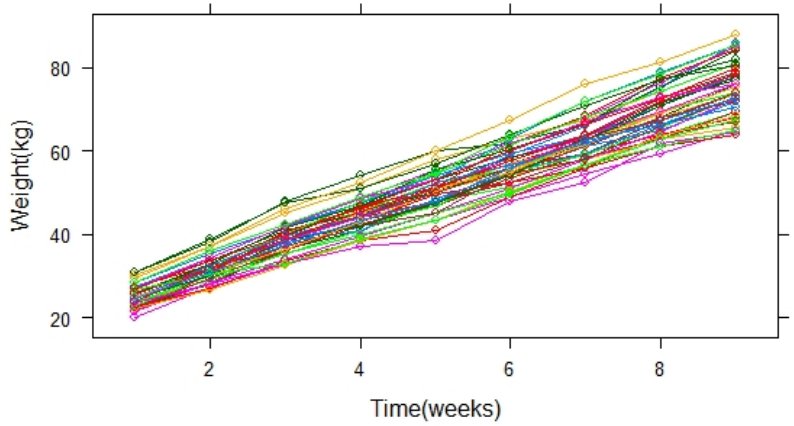
Refer to Lab5 for the R codes of graphical presentations

Example1 Bodyweights of pigs in nine successive weeks of follow-up.

A cohort study that aims to get information about the growth of the pigs.



Graphical presentation



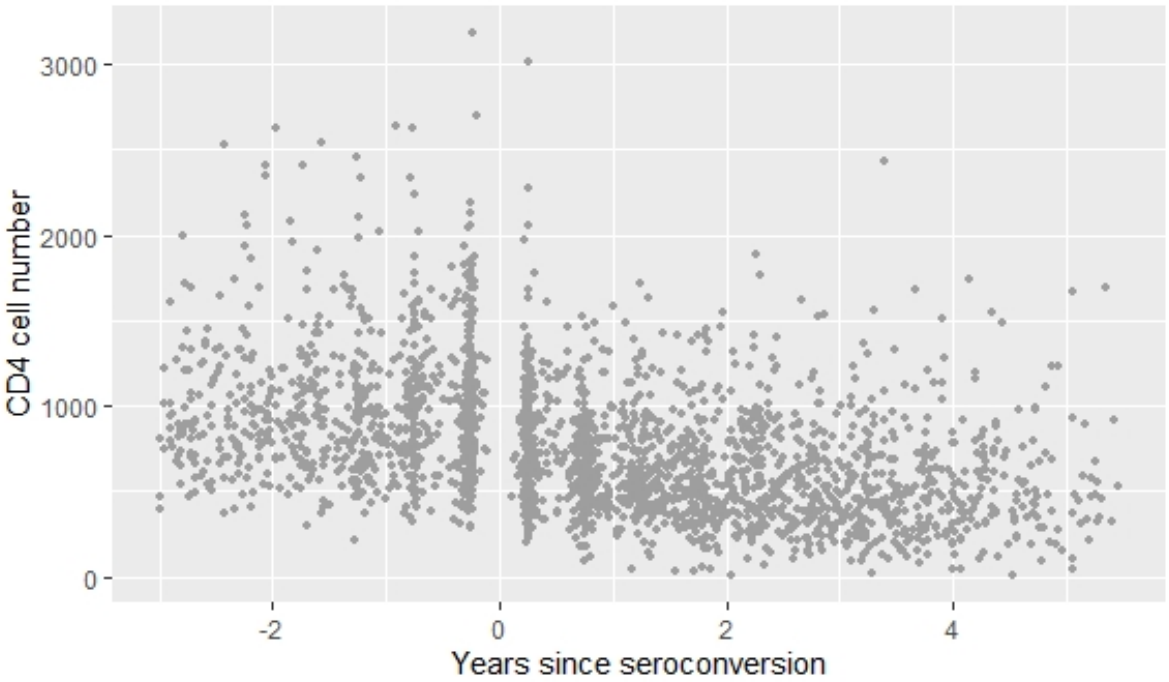
Lines connect the repeated observations for each animal.

All animals are gaining weight. Second, the pigs which are largest at the beginning of the observation period tend to be largest throughout. This phenomenon is called "tracking". Third, the spread among the animals is substantially smaller at the beginning of the study than at the end.

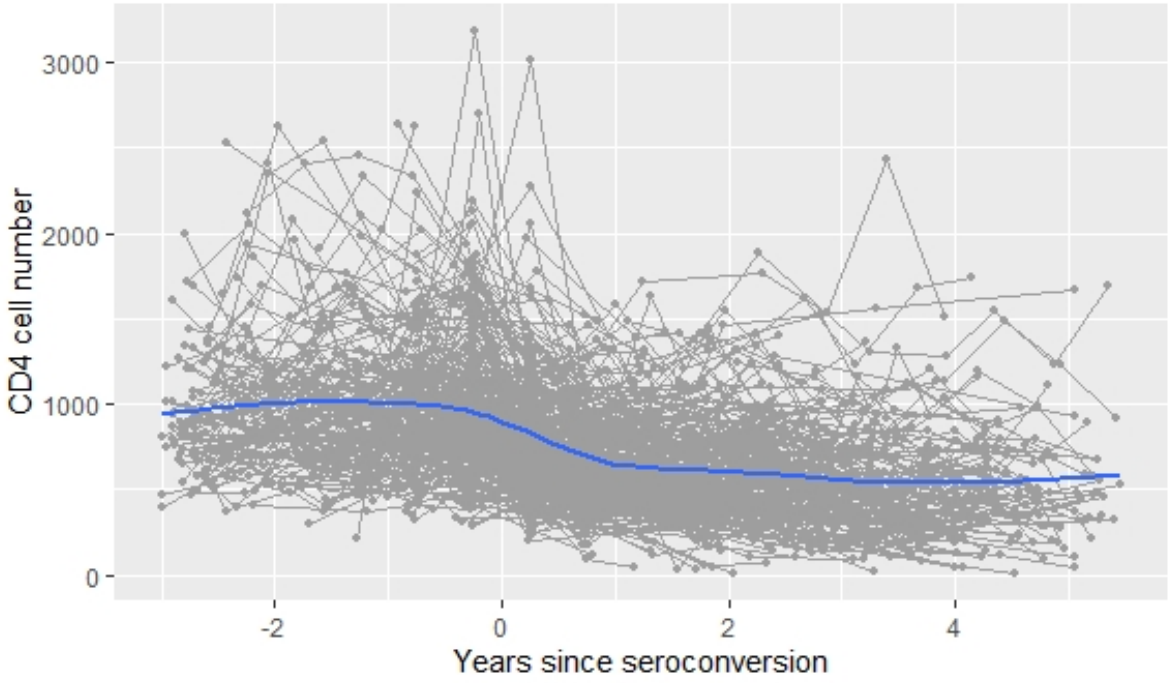
Graphical presentation

Example2 CD4 cell numbers among HIV seroconverters from the Multicenter AIDS Cohort Study (MACS). This dataset contains observations of CD4 cell counts on 369 men infected with the HIV virus.

See the trajectories of 369 men against the time variable, years since seroconversion (response against time).



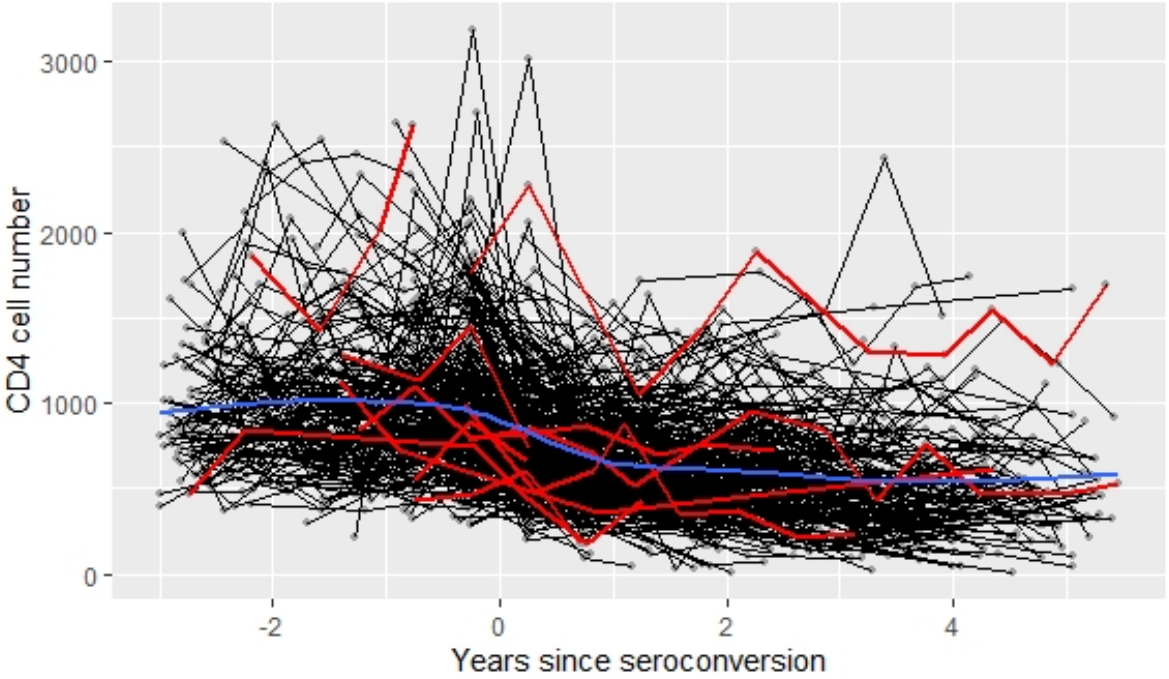
Graphical presentation



Note that each person's repeated observations are connected by straight lines.

We can see that the CD4 cell numbers are relatively constant at about 1000 cells until the time of seroconversion and then decrease afterwards. This mean pattern is highlighted in blue by the non-parametric curve.

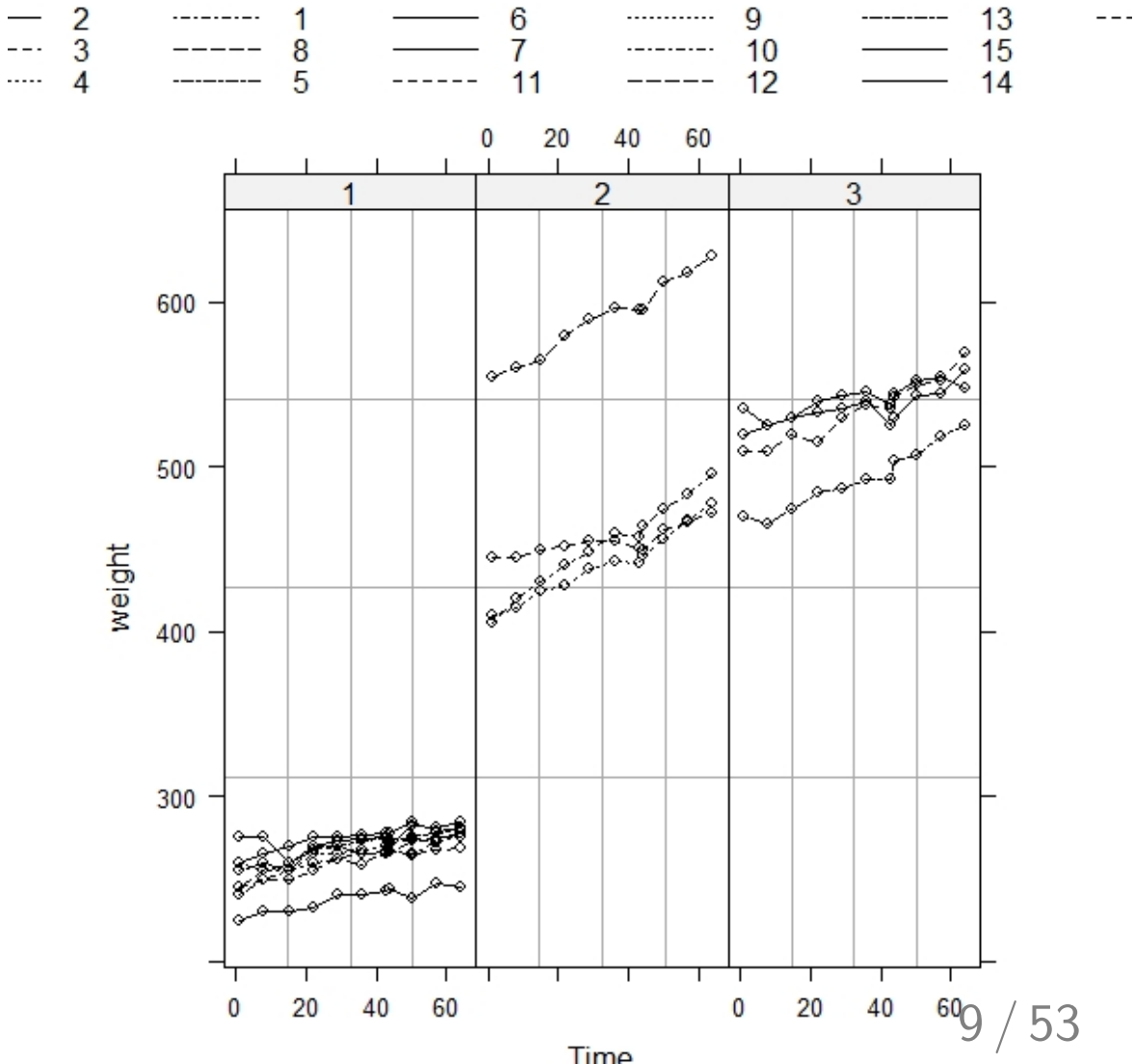
Graphical presentation



Individual's (person's) profile for a subset of data.—Attention: This is an unbalanced design.

Graphical presentation

Example3 The data are measurements of the weights of 16 rats over time. Eight of the rats were given a control diet, 4 rats were given one experimental diet (high dose, say) and 4 were given another diet (low dose, say).



Graphical presentation

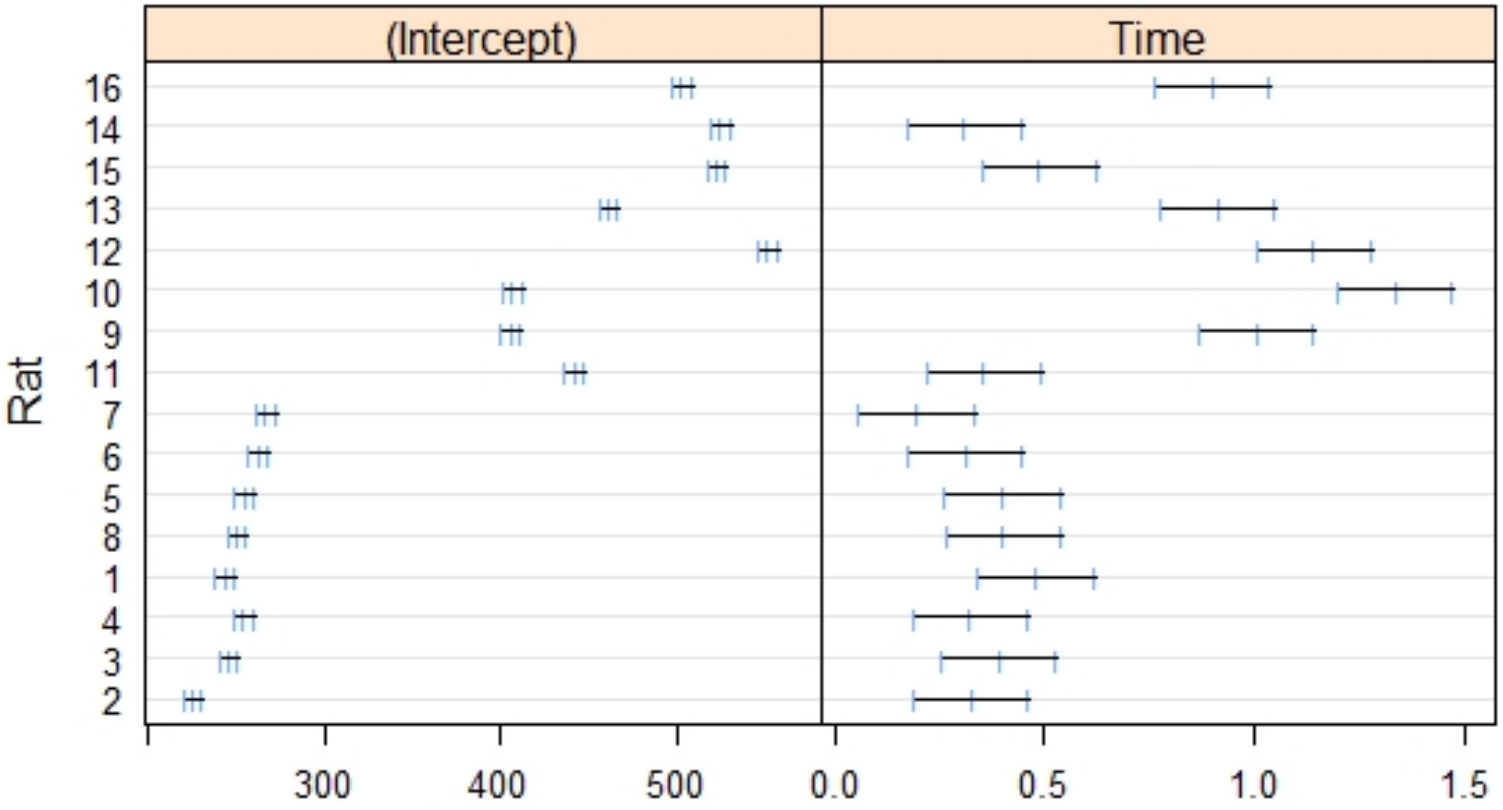
We observe that

(i) the response is approximately linear over time for all rats,
(ii) diets 2 and 3 produce heavier rats than diet 1,
(iii) growth rate for diet 1 is less that the rates for diets 2 and 3,
(iv) the variability in diets 2 and 3 is greater than in diet 1,
(v) within diets slopes are similar,
(vi) differences among rats at the beginning are preserved throughout the trial.

Graphical presentation-Example 3 cont.

We want to visualize randomness in the Rat Data.

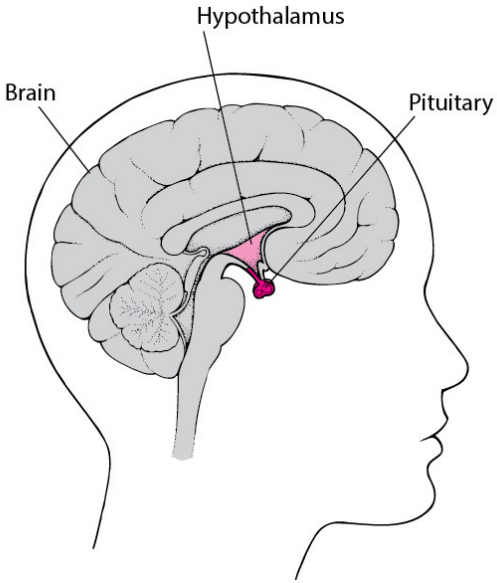
Figure: Plot of intercept and slope estimates with confidence bands

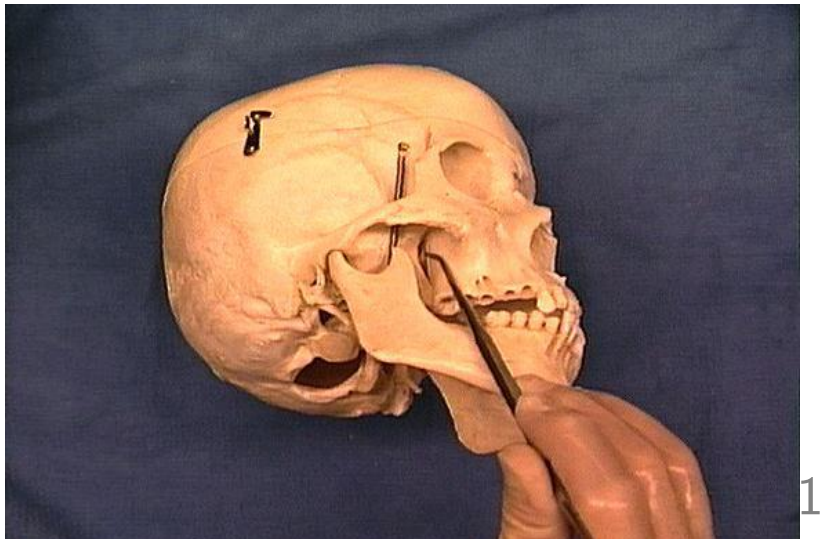


Exploratory analysis-Lab6

Does orthodontics have side-effects and faults?

Investigators at the University of North Carolina Dental School followed the growth of 27 children (16 males, 11 females) from age 8 until age 14 (four ages 8, 10, 12, 14). Every two years they measured the distance between the pituitary and the pterygomaxillary fissure, two points that are easily identified on x-ray exposures of the side of the head.





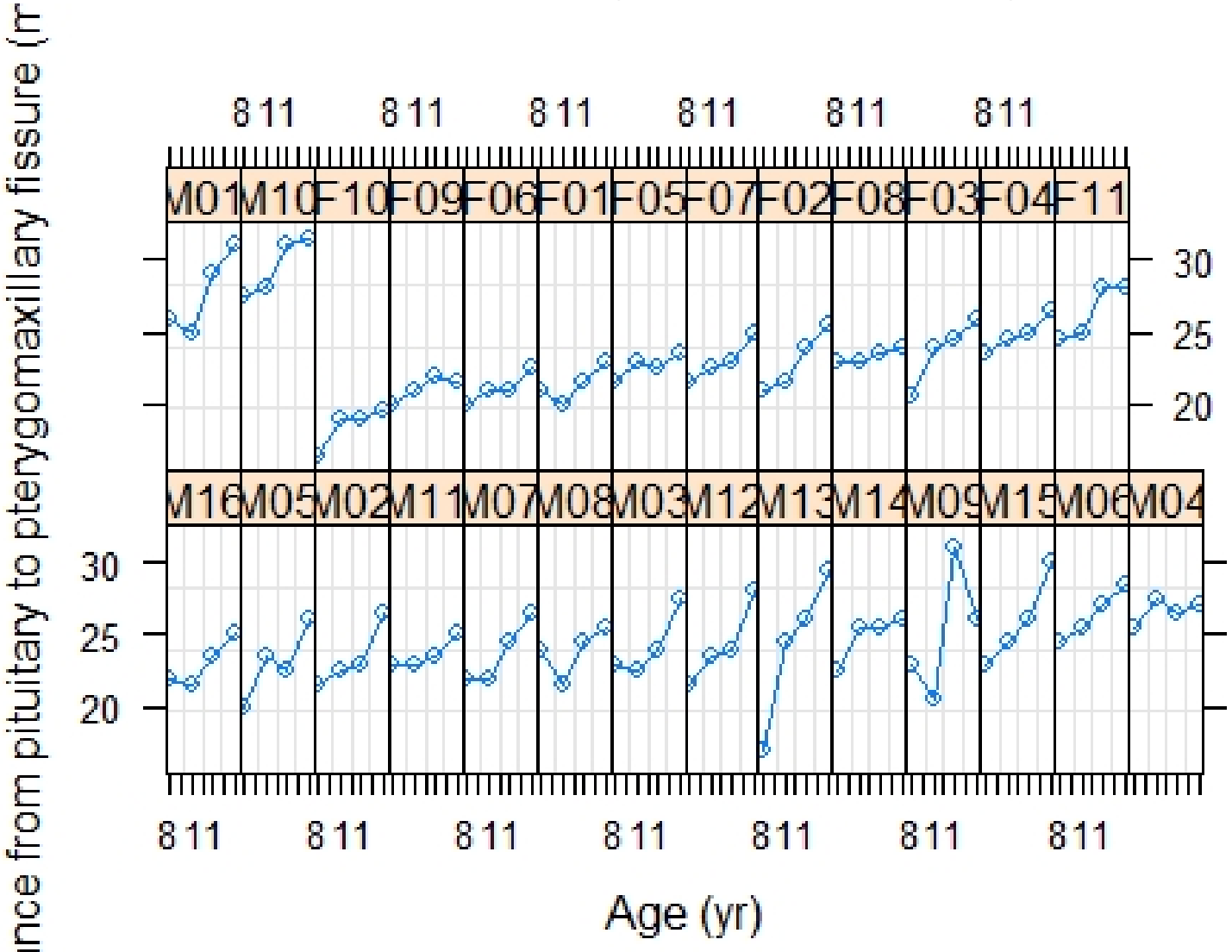
Exploratory analysis-Lab6 cont.

This data frame contains the following columns:

- **distance** a numeric vector of distances from the pituitary to the pterygomaxillary fissure (mm). These distances are measured on x-ray images of the skull.
- **age** a numeric vector of ages of the subject (yr).
- **Subject** an ordered factor indicating the subject on which the measurement was made. The levels are labeled M01 to M16 for the males and F01 to F13 for the females. The ordering is by increasing average distance within sex.
- **Sex** a factor with levels Male and Female

Exploratory analysis-Lab6 cont.

Note that it is balanced design analysis; 4 measurements for all subjects. The research question is: Is dental growth related to gender?



Exploratory analysis-Lab6 cont.

Run the R codes in Lab6 to get information about
Residual analysis and plots
Diagnostic plots for a fit
Variogram: See assignment Q1.

Exploratory analysis-Variogram-Lab6 cont.

The variogram or according to our main reference, the empirical semi-variogram is an alternative, nonparametric technique (compared to the serial correlation) which does not require the fitting of linear mixed models. It describes the association among repeated values. It is defined (Diggle et al., 2002)

$$\gamma(u) = \frac{1}{2}\mathrm{E}\left[\{Y(t) - Y(t-u)\}^2\right], \quad u \geq 0. \tag{1.1}$$

If $Y(t)$ is stationary the variogram is directly related to the autocorrelation function, $\rho(u)$, by

$$\gamma(u) = \sigma^2\{1 - \rho(u)\}$$

where σ^2 is the variance of $Y(t)$.
However, the variogram is also well-defined for a limited class of non-stationary processes for which the increments, $Y(t) - Y(t-u)$, are stationary.

Exploratory analysis-Variogram-Lab6 cont.

In the longitudinal context, the empirical counterpart of the variogram, which we call the sample variogram, is calculated from observed half-squared-differences between pairs of residuals,

$$v_{ijk} = \frac{1}{2}(r_{ij} - r_{ik})^2,$$

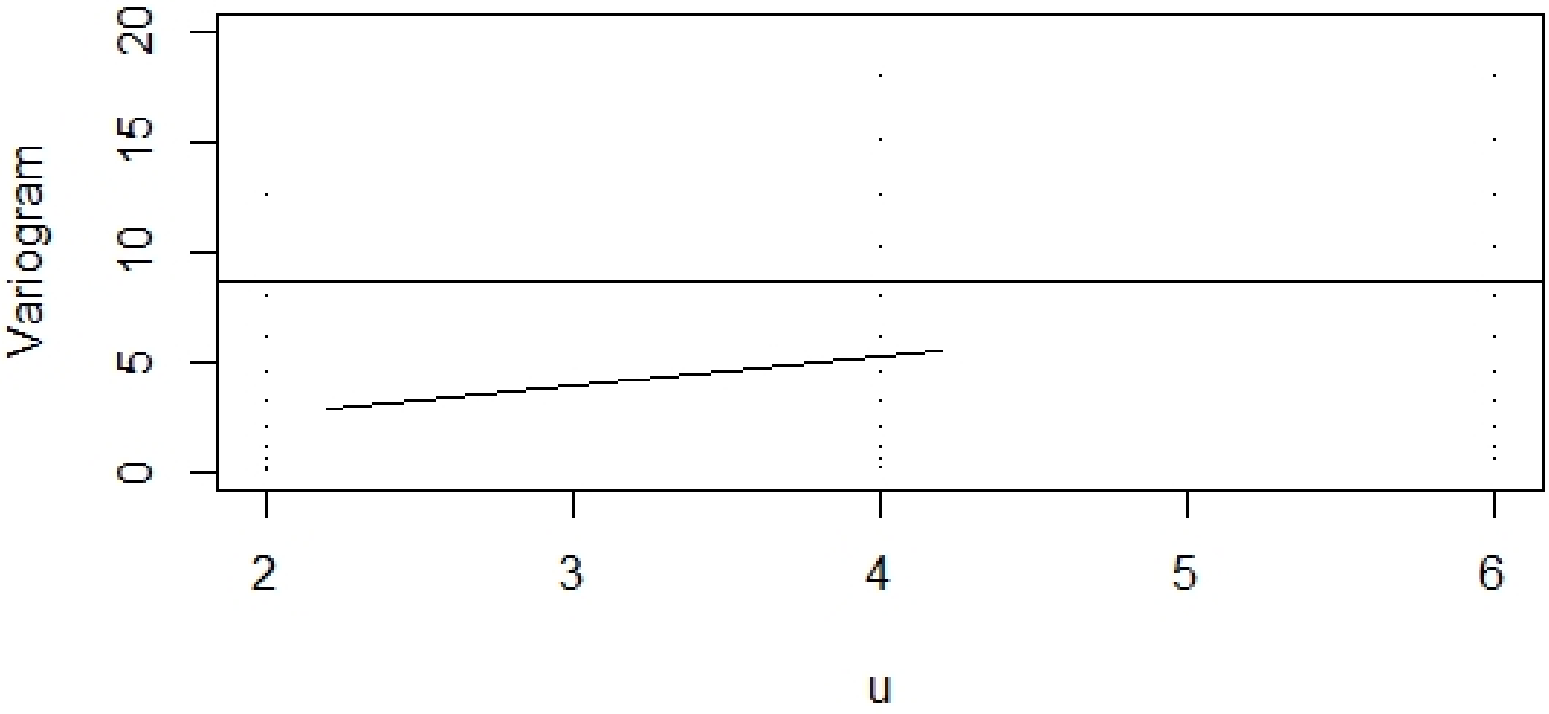
and the corresponding time-differences

$$u_{ijk} = t_{ij} - t_{ik}.$$

We can calculate the variogram for the data itself, however, for residual we need a model (Why?).

Exploratory analysis-Variogram-Lab6 cont.

Variogram for the Orthodont data as a measure of association between the distances. As the time lag increases, association decreases.



Data Layout-long format

Table: Subject is the cluster/class/group while observation is the individual/person/people.

subject	observation	response	Covariates		
1	1	y_{11}	x_{111}	$\cdots$	x_{11p}
1	2	y_{12}	x_{121}	$\cdots$	x_{12p}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$
1	n_1	y_{1n_1}	x_{1n_11}	$\cdots$	x_{1n_1p}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	1	y_{n1}	x_{n11}	$\cdots$	x_{n1p}
n	2	y_{n2}	x_{n21}	$\cdots$	x_{n2p}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$
n	n_n	y_{nn_n}	x_{nn_n1}	$\cdots$	x_{nn_np}

Data Layout

More details:

subject : $i = 1, 2, \cdots, n$

observation : $j = 1, 2, \cdots, n_i$

covariate specific index : $k = 1, 2, \cdots, p$

Elements

response : y_{ij} is the response variable at time t_{ij} for the i-th subject and j-th individual/observation

covariate : x_{ijk} is the k-th covariate at time t_{ij} for subject i, individual j

In the "wide" format, the layout of data is changed where a subject's repeated responses will be in a single row (multiple records per subject), and each response is in a separate column.

MANOVA Data Layout

What we provided was for univariate response. How about multivariate ?

Table: Data layout for the multivariate analysis used in MANOVA

subject	observation	Response			Covariates		
1	1	y_{111}	$\cdots$	y_{11q}	x_{111}	$\cdots$	x_{11p}
1	2	y_{121}	$\cdots$	y_{12q}	x_{121}	$\cdots$	x_{12p}
$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$
1	n_1	y_{1n_11}	$\cdots$	y_{1n_1q}	x_{1n_11}	$\cdots$	x_{1n_1p}
$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	1	y_{n1}	$\cdots$	y_{n1q}	x_{n11}	$\cdots$	x_{n1p}
n	2	y_{n2}	$\cdots$	y_{n2q}	x_{n21}	$\cdots$	x_{n2p}
$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$
n	n_n	y_{nn_n}	$\cdots$	y_{nn_nq}	x_{nn_n1}	$\cdots$	x_{nn_np}

Refer to Verbeke et al. (2014)[1] for a review.

[1] Verbeke, G., Fieuws, S. Molenberghs, G. (2014) The analysis of multivariate longitudinal data: A review, Stat. Meth. Med. Res., 23(1), 42-59.

An example of a bivariate response

longitudinal data on the Primary Biliary Cirrhosis sequential (PBCseq) cohort study in which 312 patients were recruited

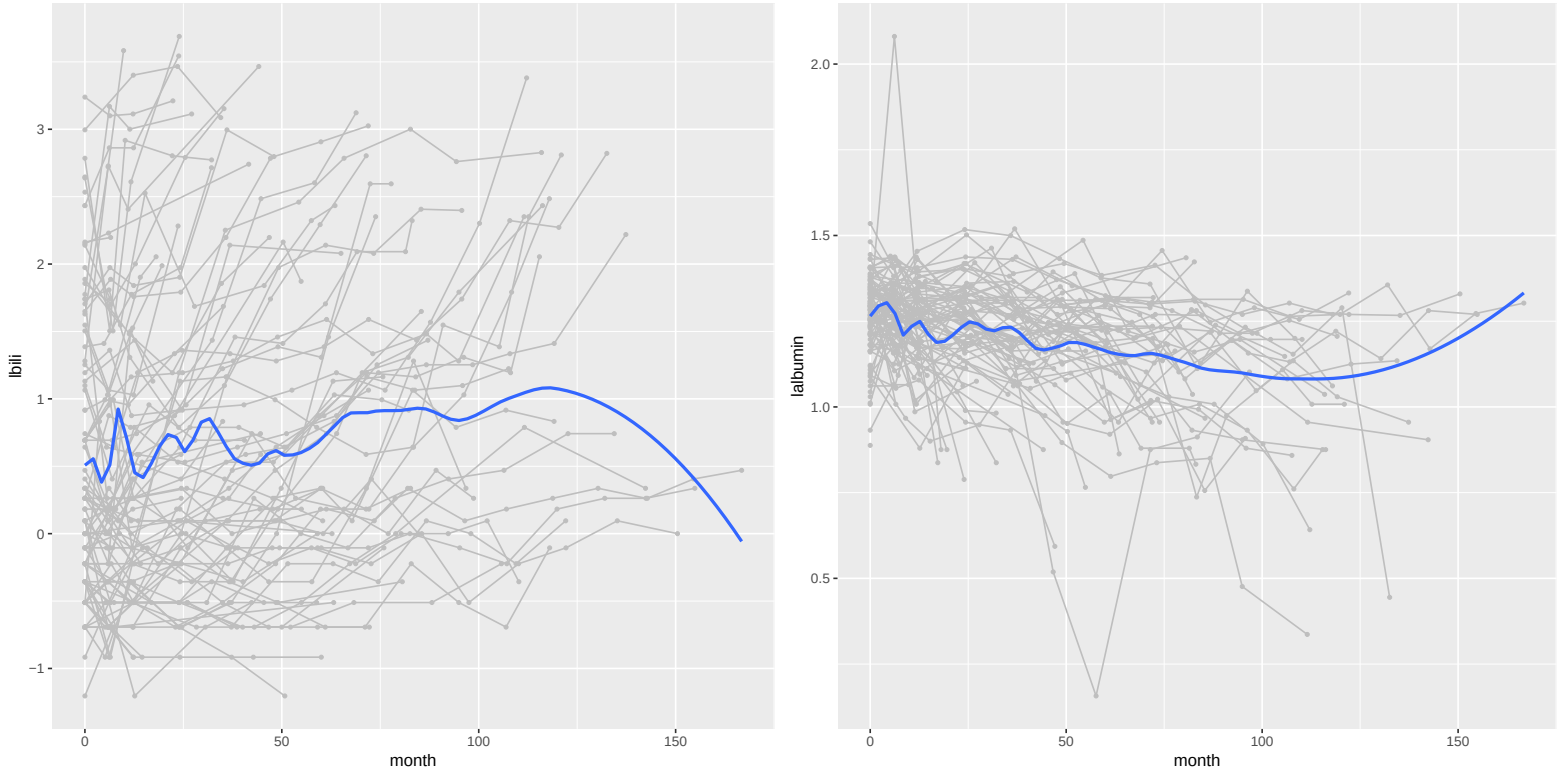


Figure: Trajectories plots for the PBCseq data. Observed evolution (in gray) of lbili and lalbumin markers for 100 randomly selected patients against time (in month). Solid (in thick blue) lines show the smoothed mean profiles of responses for all patients.

Source of Correlation

Let's put repeated measurements of subject i into one vector. If for $\boldsymbol{Y}_i = (Y_{i1}, \cdots, Y_{in_i})^\top$, $\boldsymbol{\epsilon}_i$ is the associated within-subject random error in modeling Y_i, then, which are correlated.[2]

In general, the correlation structure of the response is from three sources:

1) autocorrelation or serial correlation,
2) random effects (between-subject variability), and
3) measurement error.

[2] Remember from Session 2 the marginal models, and that observations in each cluster are correlated.

Source of Correlation-Serial correlation

Here, we list some examples

Exchangeable or Compound symmetry

$$\boldsymbol{\Sigma} = \sigma^2 \begin{bmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{bmatrix}; \mathrm{corr}(Y_{ij}; Y_{ik}) = \rho_{jk} = \rho.$$

Unstructured

$$\boldsymbol{\Sigma} = \sigma^2 \begin{bmatrix} 1 & \rho_1 & \cdots & \rho_{m-1} \\ \rho_1 & 1 & \cdots & \rho_{m-2} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{m-1} & \rho_{m-2} & \cdots & 1 \end{bmatrix}; \mathrm{corr}(Y_{ij}, Y_{ij+k}) = \rho_k.$$

Autoregressive

$$\boldsymbol{\Sigma} = \sigma^2 \begin{bmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{m-1} \\ \rho & 1 & \rho & \cdots & \rho^{m-2} \\ \rho^2 & \rho & 1 & \cdots & \rho^{m-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho^{m-1} & \rho^{m-2} & \rho^{m-3} & \cdots & 1 \end{bmatrix}; \mathrm{corr}(Y_{ij}, Y_{ik}) = \rho^{|t_{ij} - t_{ik}}$$

Approaches for LDA

- Marginal Model (remember from Session 1)

$$\mathrm{E}(\boldsymbol{Y}_i|\boldsymbol{X}_i) = \boldsymbol{X}_i\boldsymbol{\beta}, \mathrm{Var}(\boldsymbol{Y}_i) = \boldsymbol{V}_i(\boldsymbol{\alpha})$$

- Mixed-effects Model (remember from Session 1)

$$\mathrm{E}(\boldsymbol{Y}_i|\boldsymbol{X}_i, \boldsymbol{Z}_i) = \boldsymbol{X}_i\boldsymbol{\beta} + \boldsymbol{Z}_i\boldsymbol{b}, \mathrm{Var}(\boldsymbol{Y}_i) = \boldsymbol{V}_i = \boldsymbol{Z}_i\boldsymbol{D}_i\boldsymbol{Z}^\top + \boldsymbol{\Sigma}_i$$

- Transition Models (we do not discuss her): We model the response mean given the previous responses, i.e., $\mathrm{E}(Y_{ij}|Y_{ij-1}, \cdots, Y_{i1}, \boldsymbol{x}_{ij})$

A general model

Before we proceed with the LMM and marginal model, we can consider the following general model for the longitudinal data

$$Y_{ij} = \mu_{ij} + a_i + W_i(t_{ij}) + \epsilon_{ij}, \tag{2.1}$$

where

- $a_i \overset{iid}{\sim} N(0, \sigma_a^2)$ (random intercept)
- $W_i(t_{ij})$ centered continuous-time Gaussian processes modeling serial correlation, and
- $\epsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\epsilon^2)$ (measurement error)

A general model cont.

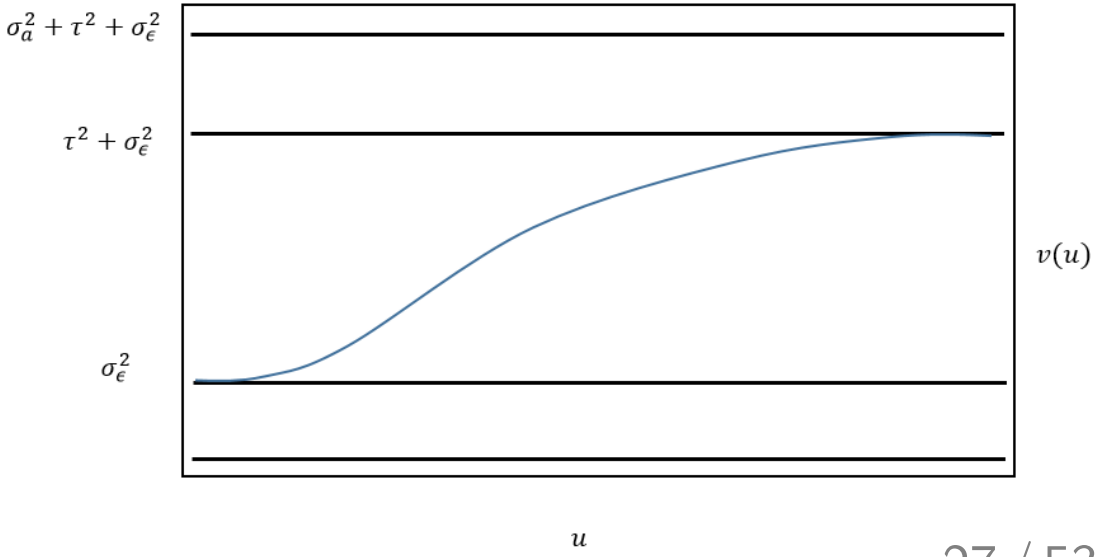
Because of independency between the components

$$\mathrm{Var}(Y_{ij}) = \sigma_a^2 + \tau^2 + \sigma_\epsilon^2. \tag{2.2}$$

If the process W_i is stationary then the variogram is directly related to the authocorrelation function $\rho(u)$, by

$$\gamma(u) = \sigma^2 + \tau^2\{1 - \rho(u)\}. \tag{2.3}$$

Remeber the variogram of Orthodont data (page 18); increases with lag (evidence of serial correlation); there is a random intercept (the variogram does not start at 0); evidence of measurement error (the variogram does not reach the total variance).



LMM

Recall from Session 2,

$$\boldsymbol{Y}_i = \boldsymbol{X}_i\boldsymbol{\beta} + \boldsymbol{Z}_i\boldsymbol{b}_i + \boldsymbol{\epsilon}_i,$$

with $\boldsymbol{Y}_i = (Y_{i1}, \cdots, Y_{in_i})^\top : n_i \times 1$, $\boldsymbol{X}_i = (\boldsymbol{x}_{i1}^\top, \cdots, \boldsymbol{x}_{in_i}^\top)^\top : n_i \times p$, $\boldsymbol{Z}_i = (\boldsymbol{z}_{i1}^\top, \cdots, \boldsymbol{z}_{in_i}^\top)^\top : n_i \times q$, $\boldsymbol{\epsilon}_i = (\epsilon_{i1}, \cdots, \epsilon_{in_i})^\top : n_i \times 1$.
Further, we assume

$$\begin{aligned} \boldsymbol{b}_i &\overset{iid}{\sim} N_q(\boldsymbol{0}, \boldsymbol{D}) \quad \text{fixed covaraince this time} \\ \boldsymbol{\epsilon}_i &\overset{iid}{\sim} N_{n_i}(\boldsymbol{0}, \boldsymbol{\Sigma}_i) \end{aligned} \tag{2.4}$$

Then, the covariance of the model is given by
$\boldsymbol{V}_i = \mathrm{Cov}(\boldsymbol{Y}_i) = \boldsymbol{Z}_i\boldsymbol{D}\boldsymbol{Z}_i^\top + \boldsymbol{\Sigma}_i$.

Example 3 cont.

Recall the rat data in Example 3 on page 9. We had 3 diets: the low dose (L_i), high does (H_i), and control does (C_i). The response was weight, and here we assume

$$Y_{ij} = (\beta_0 + b_{1i}) + (\beta_1 L_i + \beta_2 H_i + \beta_3 C_i + b_{2i}) t_{ij} + \epsilon_{ij}, \tag{2.5}$$

which can be represented as

$$Y_{ij} = \begin{cases} \beta_0 + b_{1i} + (\beta_1 + b_{2i}) t_{ij} + \epsilon_{ij} & ; \text{low does} \\ \beta_0 + b_{1i} + (\beta_2 + b_{2i}) t_{ij} + \epsilon_{ij} & ; \text{high does} \\ \beta_0 + b_{1i} + (\beta_3 + b_{2i}) t_{ij} + \epsilon_{ij} & ; \text{control} \end{cases} \tag{2.6}$$

The subject-specific profiles are assumed to be linear, with subject-specific intercepts as well as slopes. The average evolution is also linear, with different slopes for the three treatment groups, but with common intercepts. The random intercept model is obtained by $b_{2i} = 0$. What about random slope only?

Example 3 cont.

Assuming independency and common variance $\boldsymbol{\Sigma}_i = \sigma_\epsilon^2 \boldsymbol{I}_{n_i}$, the covariance function is given by

$$\begin{aligned} \mathrm{Cov}(\boldsymbol{Y}_i(t_1), \boldsymbol{Y}_i(t_2)) &= \begin{pmatrix} 1 & t_1 \end{pmatrix} \boldsymbol{D} \begin{pmatrix} 1 \\ t_2 \end{pmatrix} + \sigma_\epsilon^2, \quad \boldsymbol{D} = \begin{pmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{pmatrix} \\ &= d_{11} + d_{12}(t_1 + t_2) + d_{22} t_1 t_2 + \sigma_\epsilon^2. \end{aligned} \tag{2.7}$$

Note how the model now implies the variance function of the response to be quadratic over time, with positive curvature d_{22}.
Note in (2.7), $q = 2$, $\boldsymbol{Z}_1 = (1 \ t_1)$, $\boldsymbol{Z}_2 = (1 \ t_2)$.

Example 3 cont.

Assuming a random intercept model, $t_1 = t_2 = 0$. Thus (page 9 of Session 2),

$$\begin{aligned} \mathrm{Cov}(\boldsymbol{Y}_i(t_1), \boldsymbol{Y}_i(t_2)) &= \begin{pmatrix} 1 & 0 \end{pmatrix} \boldsymbol{D} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \sigma_\epsilon^2 \\ &= d_{11} + \sigma^2. \end{aligned} \tag{2.8}$$

Thus $\mathrm{Var}(Y_{ij}) = d_{11} + \sigma^2$ ($d_{11} = \sigma_a^2$ for reference to page 9 of Session 2). Further the interaclass correlation is equal to $\rho = \frac{d_{11}}{d_{11}+\sigma_\epsilon^2}$, between any two repeated measurements within the same rat.

Marginal model-review from Session 2

Recall the marginal model (we assume $\boldsymbol{D}_i = \boldsymbol{D}$)

$$\boldsymbol{Y}_i \sim N_{n_i}(\boldsymbol{X}_i\beta, \boldsymbol{Z}_i\boldsymbol{D}\boldsymbol{Z}_i^\top + \boldsymbol{\Sigma}_i). \tag{2.9}$$

Here, $\boldsymbol{\beta}$ is the vector of fixed-effects regression coefficients; we assume $\boldsymbol{\theta} = (\boldsymbol{\beta}^\top, \boldsymbol{\alpha}^\top)^\top$ (page 31 in Session 2), and $\boldsymbol{\alpha}$ is the vector of all components in $\boldsymbol{D}$ and $\boldsymbol{\Sigma}_i$.

REML

We first combine all $\boldsymbol{Y}_i$ into one model to get $\boldsymbol{Y} \sim N_N(\boldsymbol{X}\boldsymbol{\beta}, \boldsymbol{V}(\boldsymbol{\alpha}))$, where

$$\boldsymbol{Y} = \begin{pmatrix} \boldsymbol{Y}_1 \\ \boldsymbol{Y}_2 \\ \vdots \\ \boldsymbol{Y}_n \end{pmatrix}, \ \boldsymbol{X} = \begin{pmatrix} \boldsymbol{X}_1 \\ \boldsymbol{X}_2 \\ \vdots \\ \boldsymbol{X}_n \end{pmatrix}, \ \boldsymbol{V}(\boldsymbol{\alpha}) = \begin{pmatrix} \boldsymbol{V}_1(\boldsymbol{\alpha}) & & 0 \\ & \ddots & \\ 0 & & \boldsymbol{V}_n(\boldsymbol{\alpha}) \end{pmatrix} \tag{2.10}$$

We transform the data and define the vector of $N-1$ error contrasts as (see Eq. (4.13) in Session 2)

$$\boldsymbol{U} = \begin{pmatrix} Y_1 - Y_2 \\ Y_2 - Y_3 \\ \vdots \\ Y_{N-2} - Y_{N-1} \\ Y_{N-1} - Y_N \end{pmatrix} = \boldsymbol{A}^\top \boldsymbol{Y} \sim N_{N-p}(0, \boldsymbol{A}^\top \boldsymbol{V}(\boldsymbol{\alpha})\boldsymbol{V}) \tag{2.11}$$

REML cont.

- The MLE of $\boldsymbol{\alpha}$, based on $\boldsymbol{U}$ is called the REML estimate, and is denoted by $\hat{\boldsymbol{\alpha}}_{\mathrm{REML}}$.
- The REML estimators for $\boldsymbol{\alpha}$ and for $\boldsymbol{\beta}$ can also be found by maximizing the so-called REML likelihood function

$$L_{\mathrm{REML}}(\boldsymbol{\theta}) = \left| \sum_{i=1}^{n} \boldsymbol{X}_i^\top \boldsymbol{V}_i^{-1}(\boldsymbol{\alpha}) \boldsymbol{X}_i \right|^{-\frac{1}{2}} L_{\mathrm{LM}}(\boldsymbol{\theta}) \tag{2.12}$$

where $L_{\mathrm{LM}}(\boldsymbol{\theta})$ was given by Eq. (4.11) in Session 2.

Inference

Recall the estimate of $\boldsymbol{\beta}$

$$\hat{\boldsymbol{\beta}}(\boldsymbol{\alpha}) = \left(\sum_{i=1}^{n} \boldsymbol{X}_i \boldsymbol{V}_i^{-1}(\boldsymbol{\alpha}) \boldsymbol{X}_i \right)^{-1} \sum_{i=1}^{n} \boldsymbol{X}_i^{\top} \boldsymbol{V}_i^{-1}(\boldsymbol{\alpha}) \boldsymbol{Y}_i. \tag{2.13}$$

Also

$$\hat{\boldsymbol{\beta}}(\boldsymbol{\alpha}) | \boldsymbol{\alpha} \sim N_p \left(\boldsymbol{\beta}, \left(\sum_{i=1}^{n} \boldsymbol{X}_i \boldsymbol{V}_i^{-1}(\boldsymbol{\alpha}) \boldsymbol{X}_i \right)^{-1} \right) \tag{2.14}$$

In practice one again replaces ? by its ML or REML estimate.

Inference: approximate Wald tests

For each parameter β_j in $\boldsymbol{\beta}$, $j = 1, \cdots, p$, an approximate Wald test (also termed Z-test), as well as an associated confidence interval (CI), is obtained from approximating the distribution of $(\hat{\beta}_j - \beta_j)/\widehat{\text{s.e.}}(\hat{\beta}_j)$ by a standard univariate normal distribution. In general, for any known matrix $\boldsymbol{L}$, a test for the hypothesis

$$\mathcal{H}_o : \boldsymbol{L}\boldsymbol{\beta} = \mathbf{0} \quad \text{vs} \quad \mathcal{H}_A : \boldsymbol{L}\boldsymbol{\beta} \neq \mathbf{0} \tag{2.15}$$

immediately follows from the fact that the distribution of

$$(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top \boldsymbol{L}^\top \left[\boldsymbol{L} \left(\sum_{i=1}^{n} \boldsymbol{X}_i \boldsymbol{V}_i^{-1}(\hat{\alpha}) \boldsymbol{X}_i \right)^{-1} \boldsymbol{L}^\top \right]^{-1} \boldsymbol{L}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \tag{2.16}$$

asymptotically follows a chi-squared distribution with rank($\boldsymbol{L}$) degrees of freedom. Thus, the Wald test statistic is given by

$$\hat{\boldsymbol{\beta}}^\top \boldsymbol{L}^\top \left[\boldsymbol{L} \left(\sum_{i=1}^{n} \boldsymbol{X}_i \boldsymbol{V}_i^{-1}(\hat{\alpha}) \boldsymbol{X}_i \right)^{-1} \boldsymbol{L}^\top \right]^{-1} \boldsymbol{L}\hat{\boldsymbol{\beta}} \tag{2.17}$$

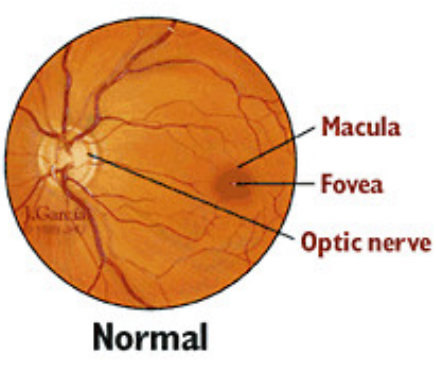
For the variance components, asymptotically, ML and REML estimates of $\boldsymbol{\alpha}$ are normally distributed with correct mean and inverse Fisher information matrix as covariance. Thus, the Wald tests can be easily obtained.

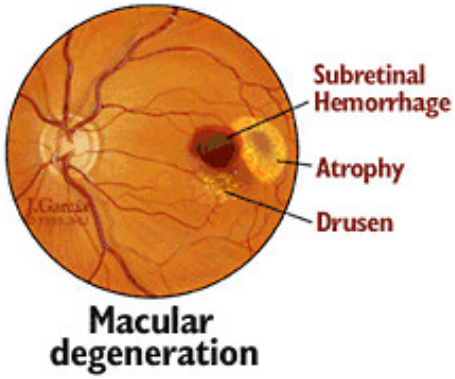
Refer to section 6 in chapter 15 of Galecki and Burzykowski (2013)[3] for a simulation. In the next slide, we will analyze a real data set.

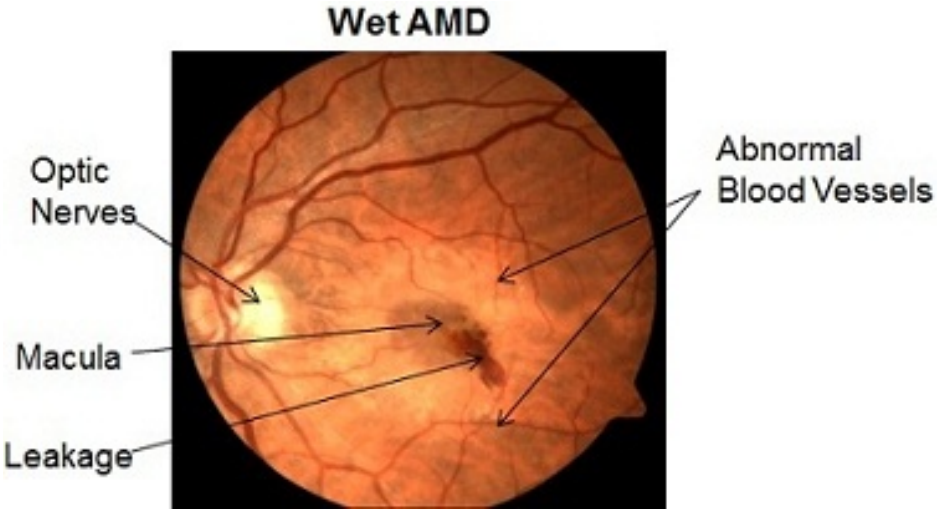
[3] Galecki, A. and Burzykowski, T. (2013) Linear Mixed-Effects Models Using R: A Step-by-Step Approach, Springer, New York.

Lab7: Analysis of ARMD data

Here, we review the analysis of the age-related macular degeneration (ARMD) data from Chapter 16 of Galecki, A. and Burzykowski, T. (2013).







Lab7 cont.

The ARMD data arise from a randomized multi-center clinical trial comparing an experimental treatment (interferon-α) versus placebo for patients diagnosed with ARMD.

We focus on the comparison between placebo and the highest dose (6 million units daily) of interferon-α. Patients with macular degeneration progressively lose vision. In the trial, visual acuity of each of 240 patients was assessed at baseline and at four post-randomization timepoints, i.e., at 4, 12, 24, and 52 weeks. Visual acuity was evaluated based on patient's ability to read lines of letters on standardized vision charts.

It follows that, for each of 240 patients, we have longitudinal data in the form of up to five visual acuity measurements collected at different, but common to all patients, time points.

Lab7 cont.

The variable subject contains patients' identifiers. Treatment identifiers are contained in the variable treat. Variables visual0, visual4, visual12, visual24, and visual52 store visual acuity measurements obtained at baseline and week 4, 12, 24, and 52, respectively. Be careful about the format of the data; see details in Lab7 R codes.

We have two formats "wide" format (armd.wide) and "long" format (armd0); (see Assignment Q5).

We consider the following model

$$\begin{aligned} \text{VISUAL}_{it} &= \beta_0 + \beta_1 \text{VISUAL0}_i + \beta_2 \text{TIME}_{it} + \beta_3 \text{TREAT}_i \\ &\quad + \beta_4 \text{TREAT}_i \times \text{TIME}_{it} + b_{0i} + \epsilon_{it} \end{aligned} \tag{2.18}$$

Lab7 cont.

The term VISUAL_{it} in (2.18) denotes the value of visual acuity measured for patient i ($i = 1, \cdots, 240$) at time t ($t = 1, 2, 3, 4$, corresponding to values of 4, 12, 24, and 52 weeks, respectively). In the fixed-effects part of the model, VISUAL0_i is the value of visual acuity measured at baseline; TIME_{it} is the time of the measurement, corresponding to t; TREAT_i is the treatment indicator, equal 1 for the active group and 0 otherwise; and $\text{TREAT}_i \times \text{TIME}_{it}$ is their interaction. The parameter β_0 is an overall intercept, β_1 describes the change in the mean visual acuity due to a unit increase in visual acuity at baseline, β_2 describes the change due to a one week change in time, β_3 gives an overall treatment effect, and β_4 describes the additional change due to a one week change in time for patients treated with the active treatment. Further, b_{0i} is a patient-specific random intercept, assumed to be normally distributed with mean 0 and variance d_{11}, while ϵ_{it} is a residual random error assumed to be normally distributed with mean 0 and variance σ_ϵ^2.

Lab7 cont.-matrix notation

In matrix notation, the model for the subject i with a complete set of four visual acuity measurements is expressed as follows:

$$\boldsymbol{Y}_i = \begin{pmatrix} \text{VISUAL}_{i1} \\ \text{VISUAL}_{i2} \\ \text{VISUAL}_{i3} \\ \text{VISUAL}_{i4} \end{pmatrix}, \ \boldsymbol{Z}_i = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \ \boldsymbol{\epsilon}_i = \begin{pmatrix} \epsilon_{i1} \\ \epsilon_{i2} \\ \epsilon_{i3} \\ \epsilon_{i4} \end{pmatrix}, \ \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{pmatrix}$$

$$\boldsymbol{X}_i = \underbrace{\left(\begin{array}{c|c|c|c|c} 1 & \text{VISUAL0}_i & 4 & \text{TREAT}_i & 4 \times \text{TREAT}_i \\ 1 & \text{VISUAL0}_i & 12 & \text{TREAT}_i & 12 \times \text{TREAT}_i \\ 1 & \text{VISUAL0}_i & 24 & \text{TREAT}_i & 24 \times \text{TREAT}_i \\ 1 & \text{VISUAL0}_i & 52 & \text{TREAT}_i & 52 \times \text{TREAT}_i \end{array}\right)}_{\text{intercept—VISUAL0-TIEM—TREAT—TIME*TREAT}}, \ \boldsymbol{b}_i = b_{0i}$$

$$\boldsymbol{D} = d_{11}, \ \boldsymbol{\Sigma}_i = \sigma_\epsilon^2 \boldsymbol{I}_4. \tag{2.19}$$

$$\boldsymbol{Y}_i = \boldsymbol{X}_i \boldsymbol{\beta} + \boldsymbol{Z}_i \boldsymbol{b}_i + \boldsymbol{\epsilon}_i$$

Lab7 cont.

Based on (2.19), the covariance of the model is given by (see page 28)

$$
\begin{aligned}
\boldsymbol{V}_i &= \boldsymbol{Z}_i \boldsymbol{D} \boldsymbol{Z}_i^\top + \boldsymbol{\Sigma}_i \\
&= \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} d_{11} \begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix} + \begin{pmatrix} \sigma_\epsilon^2 & 0 & 0 & 0 \\ 0 & \sigma_\epsilon^2 & 0 & 0 \\ 0 & 0 & \sigma_\epsilon^2 & 0 \\ 0 & 0 & 0 & \sigma_\epsilon^2 \end{pmatrix} \\
&= \begin{pmatrix} \sigma_\epsilon^2 + d_{11} & d_{11} & d_{11} & d_{11} \\ d_{11} & \sigma_\epsilon^2 + d_{11} & d_{11} & d_{11} \\ d_{11} & d_{11} & \sigma_\epsilon^2 + d_{11} & d_{11} \\ d_{11} & d_{11} & d_{11} & \sigma_\epsilon^2 + d_{11} \end{pmatrix}. \qquad (2.20)
\end{aligned}
$$

Thus, the marginal covariance structure is exchangeable (see page 24) with a common correlation equal to $\rho = d_{11}/(\sigma_\epsilon^2 + d_{11})$.
Note that, because the variance component d_{11} is constrained to be non-negative, ρ forced to be nonnegative.

Lab7 cont.

From the obtained results, we have (standard deviations for the variance components)

$$\sqrt{d_{11}} = 8.98, \quad \sigma_\epsilon = 8.63. \tag{2.21}$$

Further, the Log-REML value is equal to -3288.986.
Also $\hat{\boldsymbol{\Sigma}}_i = \hat{\sigma}_\epsilon \boldsymbol{I}_4 = 74.43 \boldsymbol{I}_4$.

Lab7-1: Heteroscedastic variance-covariance structure

Consider the model (2.18), where we take the following structure (model 2, say)

$$\boldsymbol{\Sigma}_i = \sigma_\epsilon^2 \begin{pmatrix} (\text{TIME}_{i1})^{2\delta} & 0 & 0 & 0 \\ 0 & (\text{TIME}_{i2})^{2\delta} & 0 & 0 \\ 0 & 0 & (\text{TIME}_{i3})^{2\delta} & 0 \\ 0 & 0 & 0 & (\text{TIME}_{i4})^{2\delta} \end{pmatrix}. \tag{2.22}$$

Thus

$$\boldsymbol{V}_i = \begin{pmatrix} \sigma_1^2 + d_{11} & d_{11} & d_{11} & d_{11} \\ d_{11} & \sigma_2^2 + d_{11} & d_{11} & d_{11} \\ d_{11} & d_{11} & \sigma_3^2 + d_{11} & d_{11} \\ d_{11} & d_{11} & d_{11} & \sigma_4^2 + d_{11} \end{pmatrix}, \tag{2.23}$$

where $\sigma_t^2 = \sigma_\epsilon^2 (\text{TIME}_{it})^{2\delta}$.

Lab7-1 cont.

From the obtained results,

$$\begin{aligned} \hat{\sigma}_\epsilon &= 3.607 \\ \hat{\delta} &= 0.314 \\ \sqrt{\hat{d}_{11}} &= 7.706 \end{aligned}$$

Note that the estimated variance of random intercepts is equal to $\hat{d}_{11} = 7.706^2 = 59.38$. Note that it is smaller than the value of $8.98^2 = 80.64$, obtained for model 1. This is expected, because, by allowing for heteroscedastic residual random errors, a larger part of the total variability is explained by the residual variances.

Lab 7-1: Diagnostic plots

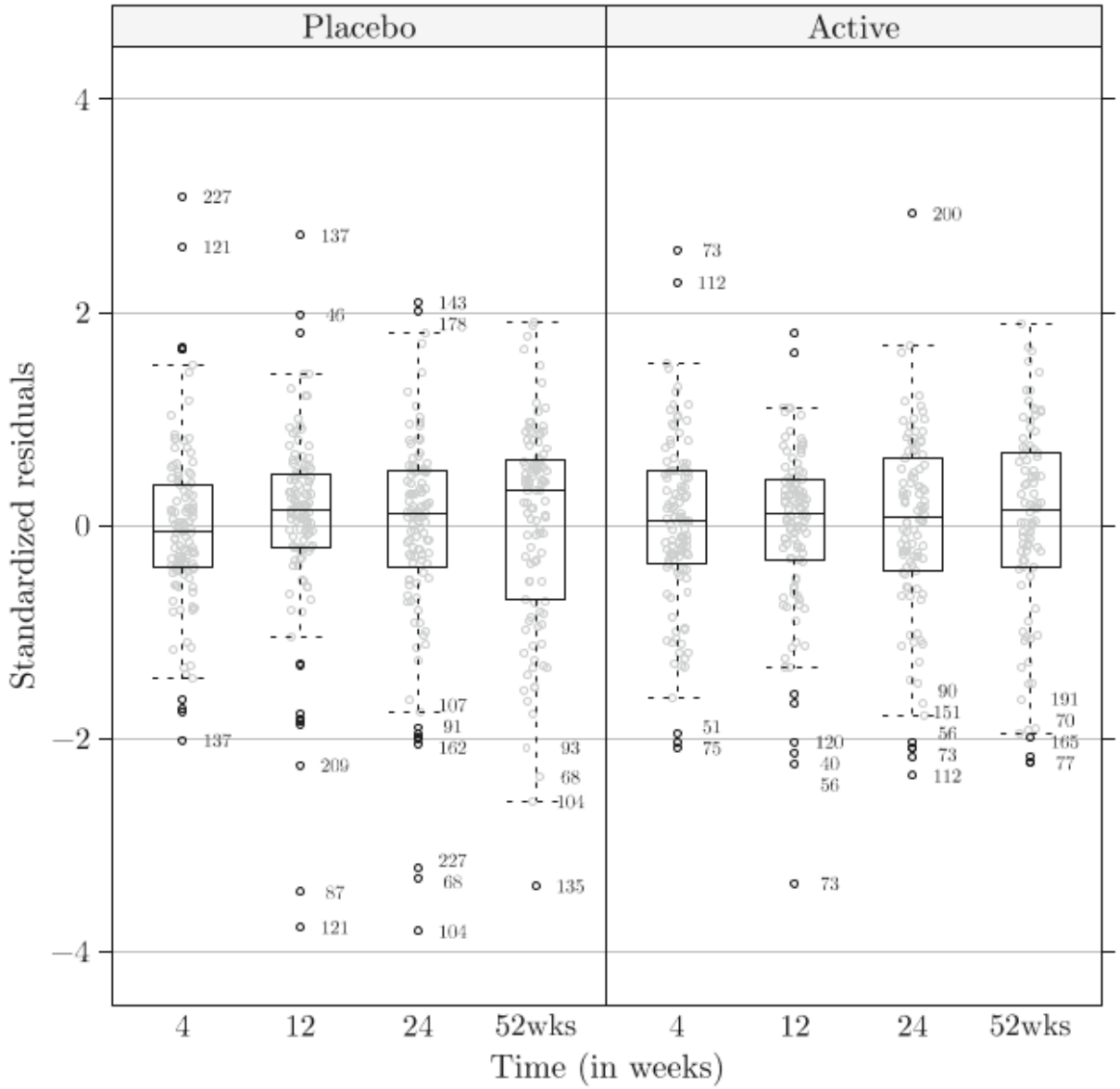


Figure 16.2 from Galecki and Burzykowski (2013)

We can see an evaluation of the distribution of the conditional Pearson residuals for each time point and treatment group. Despite standardization, the variability of the residuals seems to vary. The plot reveals also a number of outliers, i.e., residuals larger, in absolute value, than the 97.5th percentile of the standard normal distribution (they have been labeled in the plot by the corresponding observation number).

Lab 7-1: Outliers

```
> id <- 0.05                                  # Argument for qnorm()
> outliers.idx <-
+  within(armd,
+         {
+         resid.p <- resid(fm16.2, type = "pearson") # Pearson resids.
+         idx <- abs(resid.p) > -qnorm(id/2)         # Indicator vector
+         })
> outliers  <- subset(outliers.idx, idx)      # Data with outliers
> nrow(outliers)                              # Number of outliers
  [1] 38
> outliers$subject                            # IDs of outliers
   [1] 40  46  51  56  56  68  68  70  73  73  73  75  77  87  90
  [16] 91  93  104 104 107 112 112 120 121 121 135 137 137 143 151
  [31] 162 165 178 191 200 209 227 227
  234 Levels: 1 2 3 4 6 7 8 9 10 11 12 13 14 15 16 17 18 19... 240
```

Lab 7-1 cont.

Observed and predicted values of the visual acuity measurements for selected patients.

- The predicted population means, shown in the plot, decrease linearly in time.
- For some patients, the so-obtained predicted individual profiles strongly deviate from the observed ones. For instance, for the subjects 4 and 15, the predicted individual patterns suggest a decrease of visual acuity over time, while the observed values actually increase over time.

Assignments

Q1 : Write an R code to compute the variogram for CD4 data and display the graph. Analyze the output.

Q2 : List some advantages and disadvantages of longitudinal data analysis.

Q3 : What is Toeplitz matrix structure for the covariance/correlation matrix? Provide the code of a simple example in R where you use this structure in any model you would like. I must be able to run your code and see the result.

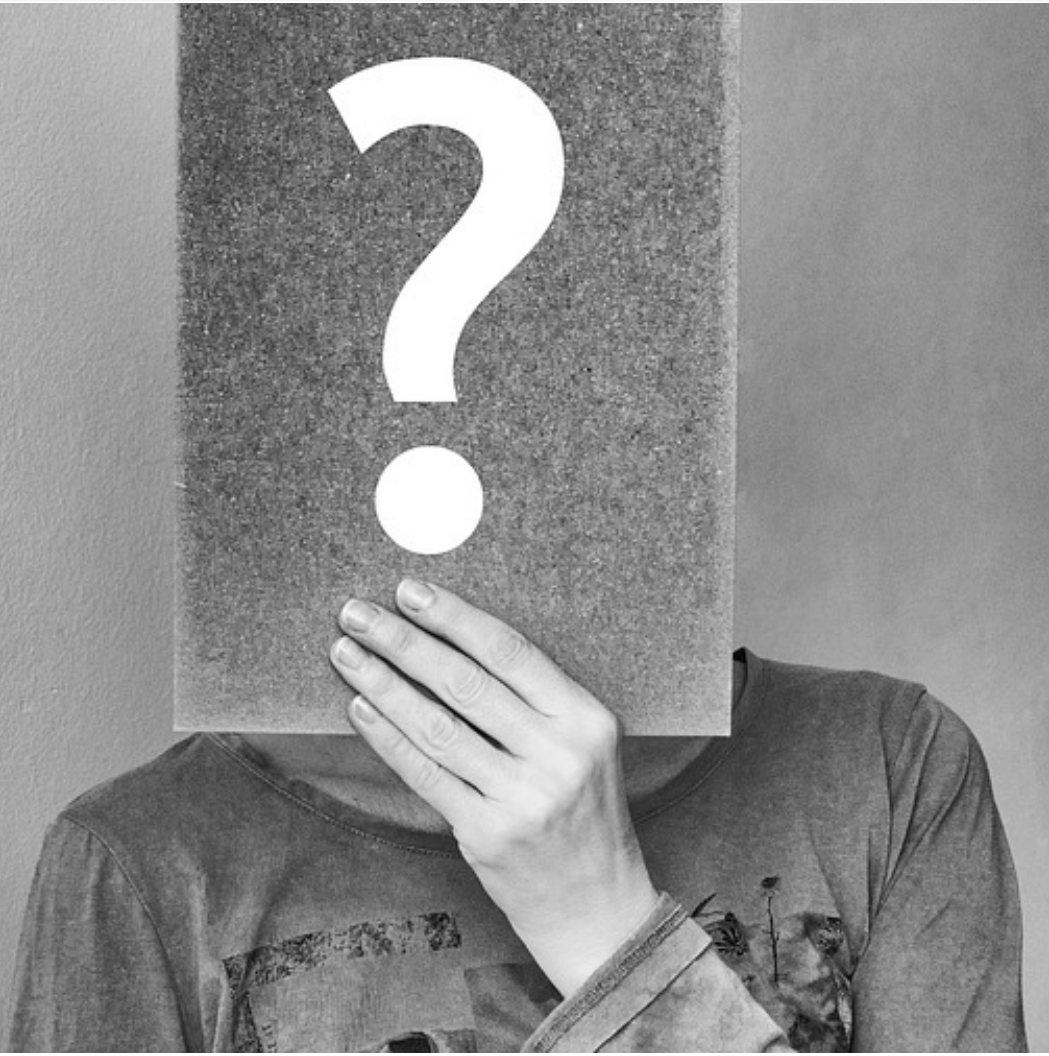
Q4 : Verify (2.14).

Q5 : What are "wide" and "long" formats in longitudinal data analysis? Provide an example (real data or synthetic).

Q6 : Write an R code that can compute $\boldsymbol{D}$, $\boldsymbol{\Sigma}_i$ and $\boldsymbol{V}_i$ for Lab7-1.

Assignments

Q7 : Consider the following model and do the whole analysis described in Labs 7 and 7-1.

$$\begin{aligned} \text{VISUAL}_{it} &= \beta_0 + \beta_1 \text{VISUAL0}_i + \beta_2 \text{TIME}_{it} + \beta_3 \text{TREAT}_i \\ &\quad + \beta_4 \text{TREAT}_i \times \text{TIME}_{it} \\ &\quad + b_{0i} + b_{2i} \text{TIME}_{it} + \epsilon_{it}. \end{aligned} \tag{2.24}$$

Thank you for your attention.